SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA37- SOFTWARE TESTING

EXP:8

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Requirement Analysis and Positive Test Scenarios and Test Cases for a Banking Application

Aim

To study the requirements of a Banking Application, and to prepare positive test scenarios and test cases that confirm the application works correctly for valid inputs and normal user actions.

Problem Statement

An internet banking application lets customers check balances, transfer money, pay bills and download statements. It handles money and confidential data, so a defect can cause financial loss, security breaches or legal problems for the bank.

Testing is needed to check that the application:

- Customers can log in securely and complete banking transactions correctly.
- Invalid inputs and unauthorized actions are rejected safely.
- Limits, balance checks and security rules follow the requirements.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	The customer logs in with a user ID and password, followed by an OTP (two-factor authentication).
FR2	The account is locked after 3 consecutive wrong password attempts.
FR3	The customer can view the account balance and a mini statement of the last 5 transactions.
FR4	The customer can add a beneficiary with name, account number and a valid IFSC code, verified by OTP.
FR5	Fund transfer is allowed only to an added beneficiary. The minimum is ₹1 and the daily limit is ₹1,00,000.
FR6	A transfer must not exceed the available balance.
FR7	Every transaction gives a reference number, a receipt and an SMS alert.
FR8	The customer can pay utility bills and recharge mobile numbers.
FR9	The customer can download a statement for a chosen period as a PDF. The from date must not be after the to date.
FR10	The customer can change the password. The new password must differ from the last 3 passwords and be strong.
FR11	The session ends after 5 minutes of inactivity, and the customer can log out securely.
FR12	Only logged-in customers can access account pages.

2 Software and environment requirements

Item	Details
Application	Web banking portal (Java Spring Boot back end, Angular front end)
Database	Oracle / MySQL (accounts, beneficiaries, transactions)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, Postman, bug tracker (Jira)
Test data	Savings and current accounts with different balances, valid and invalid IFSC codes, beneficiaries, sandbox OTP service

Positive Test Scenarios

A positive test scenario checks that a feature works as expected when the user gives valid input and follows the normal flow.

Scenario ID	Module	Test Scenario	Requirement
BK-PS01	Login	Log in with valid credentials and OTP	FR1
BK-PS02	Account	View the account balance	FR3
BK-PS03	Account	View the mini statement	FR3
BK-PS04	Beneficiary	Add a beneficiary with valid details	FR4
BK-PS05	Transfer	Transfer an amount to an added beneficiary	FR5, FR6, FR7
BK-PS06	Transfer	Transfer exactly the daily limit	FR5
BK-PS07	Transfer	Transfer the minimum amount	FR5
BK-PS08	Alerts	Receipt and SMS alert after a transaction	FR7
BK-PS09	Bill payment	Pay an electricity bill	FR8
BK-PS10	Recharge	Recharge a mobile number	FR8
BK-PS11	Statement	Download a statement for a valid period	FR9
BK-PS12	Password	Change the password with valid input	FR10
BK-PS13	Logout	Log out securely	FR11

Test Cases

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
BK-PT01	BK-PS01	1. Open the login page 2. Enter user ID and password 3. Click Login 4. Enter the OTP received	User ID: 100234; Password: Bank@2026; correct OTP	Customer is logged in and the dashboard is displayed.
BK-PT02	BK-PS02	1. Log in 2. Click Account Summary	Savings account with ₹25,000	Correct available balance is displayed.
BK-PT03	BK-PS03	1. Log in 2. Open Mini Statement	Account with 5 or more transactions	The last 5 transactions are listed with date, amount and balance.
BK-PT04	BK-PS04	1. Open Add Beneficiary 2. Enter name, account number and IFSC 3. Submit 4. Enter the OTP	Name: Anand R; A/c 123456789012; IFSC SBIN0001234	Beneficiary is added and appears in the list.
BK-PT05	BK-PS05	1. Open Fund Transfer 2. Select the beneficiary 3. Enter the amount 4. Confirm with OTP	₹5,000	Transfer succeeds, the balance is reduced by ₹5,000 and a reference number is shown.
BK-PT06	BK-PS06	1. Select a beneficiary 2. Enter the maximum limit amount 3. Confirm	₹1,00,000 with balance ₹1,50,000	Transfer is accepted at the boundary value.
BK-PT07	BK-PS07	1. Select a beneficiary 2. Enter the minimum amount 3. Confirm	₹1	Transfer is accepted at the lower boundary.
BK-PT08	BK-PS08	1. Complete a transfer 2. View the receipt 3. Check the registered mobile	Transfer of ₹2,000	Receipt has the reference number, amount and date. An SMS alert is received.
BK-PT09	BK-PS09	1. Open Bill Payments 2. Select the biller 3. Enter the consumer number and amount 4. Confirm	Consumer no. 5566778899; ₹1,200	Payment is successful and the account is debited.
BK-PT10	BK-PS10	1. Open Mobile Recharge 2. Enter the number and plan 3. Confirm	9876543210; ₹299 plan	Recharge is successful and a confirmation is shown.
BK-PT11	BK-PS11	1. Open Statements 2. Select from and to dates 3. Click Download PDF	01-10-2026 to 31-10-2026	PDF downloads and contains all transactions of the period.

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
BK-PT12	BK-PS12	1. Open Change Password 2. Enter old and a new strong password 3. Submit	Old: Bank@2026; New: Secure#2027	Password is changed and the user can log in with the new one.
BK-PT13	BK-PS13	1. Click Logout 2. Press the browser Back button	Active session	Session ends, the login page is shown and Back does not restore the session.

Result / Conclusion

Result: 13 positive test scenarios and 13 test cases were prepared for the Banking Application, covering 10 of the 12 functional requirements (no positive scenario for FR2, FR12). The test cases by module are:

Module	Test Scenarios	Test Cases
Login	1	1
Account	2	2
Beneficiary	1	1
Transfer	3	3
Alerts	1	1
Bill payment	1	1
Recharge	1	1
Statement	1	1
Password	1	1
Logout	1	1
Total	13	13

Conclusion: The positive test cases cover the main user flows of the Banking Application, from the first action to the final result. When they are executed on a build, they confirm that the valid-input behaviour matches the specification. They should be run first in every test cycle and kept in the regression suite, with the negative test cases added to check invalid inputs.